

EXPERIMENT NO. 5

Largest Element with Least Time Complexity

Programming Language: C

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Aim

To write a C program that finds the maximum element of an array using a single linear scan, which is the least possible time complexity.

Question

Write a program for the below test cases with the least time complexity.

Test Cases

Input: {1, 2, 3, 4, 5}

Expected Output: 5

Input: {7, 7, 7, 7, 7}

Expected Output: 7

Input: {-10, 2, 3, -4, 5}

Expected Output: 5

Algorithm

Step 1: Start.

Step 2: Pass the array nums and its size to the function findMax.

Step 3: Assume the first element is the maximum: $\text{max} = \text{nums}[0]$.

Step 4: Traverse the array from index 1 to size - 1; whenever $\text{nums}[i] > \text{max}$, set $\text{max} = \text{nums}[i]$.

Step 5: Return max and print it for each test case.

Step 6: Stop.

Program (C)

```
#include <stdio.h>

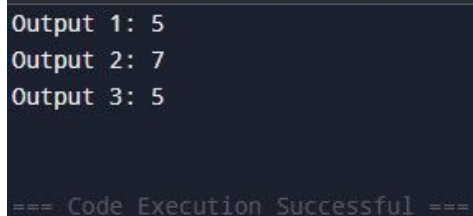
int findMax(int* nums, int size) {
    int max = nums[0];
    for (int i = 1; i < size; i++) {
        if (nums[i] > max) {
            max = nums[i];
        }
    }
    return max;
}

int main() {
    int case1[] = {1, 2, 3, 4, 5};
    int case2[] = {7, 7, 7, 7, 7};
    int case3[] = {-10, 2, 3, -4, 5};

    printf("Output 1: %d\n", findMax(case1, 5)); // Expected: 5
    printf("Output 2: %d\n", findMax(case2, 5)); // Expected: 7
    printf("Output 3: %d\n", findMax(case3, 5)); // Expected: 5

    return 0;
}
```

Output

A screenshot of a terminal window with a dark background. It displays the output of the program: "Output 1: 5", "Output 2: 7", and "Output 3: 5". At the bottom, it shows "=== Code Execution Successful ===".

```
Output 1: 5
Output 2: 7
Output 3: 5

=== Code Execution Successful ===
```

Complexity Analysis

Time Complexity: $O(n)$

Space Complexity: $O(1)$

Result

Thus the program was executed successfully and the output was verified.